Evolving a Set of Competing Economic-School Prompts: NSGA-II with LLM-Guided Mutation

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November 2026

Abstract

Multi-perspective LLM agents are useful only if each agent stays faithful to its perspective and the agents differ from one another. Prompt-optimisation work usually tunes one prompt against one score, so the trade-off between fidelity, mutual distinctness and prompt length is not searched directly. We cast the design of three economic-school agents (Keynesian, Austrian, Monetarist) as a constrained multi-objective problem over a *set* of prompts and search it with NSGA-II, using an LLM as a guided mutation operator. Each objective is measured on synthetic macroeconomic events through a disk-cached model backend. In a reduced-budget study (population 6, four generations, three seeds per method, 2,743 model calls, about USD 20 estimated), NSGA-II with LLM-guided mutation and a single-objective weighted-sum GA reach a hypervolume of about 0.12, against about 0.098 for random search and for NSGA-II with rule-based mutation only. On ten held-out events, the selected prompt sets are 20–30% shorter than the hand-written baseline with similar fidelity and disagreement, but no method improves significantly on the baseline after Holm correction, and random search gains almost as much fidelity as the evolutionary methods. We report these results as they are and discuss what the budget can and cannot establish.

Keywords: evolutionary prompt optimisation, multi-objective evolutionary algorithms, NSGA-II, LLM-guided mutation, multi-agent diversity, quality-diversity.

1 Introduction

1.1 Problem and motivation

Systems that combine several LLM agents, each speaking from a different perspective, are increasingly used for analysis, forecasting and debate (Baltaji, Hemmatian, and Varshney 2024; Takano, Hirano, and Nakagawa 2025). Their value rests on two properties that pull in opposite directions. Each agent must stay *faithful* to its perspective, and the agents must be *distinct* from one another, or the ensemble collapses into one voice. Recent evidence points to exactly this failure: debate and ensemble systems lose diversity through conformity and majority pressure (H. Wu, Li, and Li 2025; Zhu et al. 2026), and aligned models produce homogeneous outputs across many open-ended queries (Jiang et al. 2025). Prompts are the cheapest lever for setting perspective, but they are usually written by hand.

Evolutionary prompt optimisation can search prompts automatically, with an LLM supplying the variation operators (Fernando et al. 2024; Guo et al. 2024; Lange, Tian, and Tang 2024). These methods, however, optimise one prompt against one scalar score. None of them evolves a *set* of prompts whose members must be good individually and different from each other, and none exposes the trade-off between fidelity, distinctness and cost (prompt length). Multi-objective prompt search has started to appear (Zhao et al. 2025; Câmara et al. 2025). We found no such work for multi-agent sets.

A preliminary, unpublished result from the author's MPhil project, whose harness this project borrows for its measurements, points the same way. In a 240-request disagreement matrix over six economic frameworks (synthetic events, generator seed 11, the same model identifier `qwen3.8-max`), the three trained schools disagreed about twice as much as one school disagreed with itself, while New Keynesian duplicated Keynesian within the noise floor. This evidence is used here only as motivation; none of its events or outputs enter the experiments below.

1.2 Research questions

- **RQ1.** Can NSGA-II over prompt sets, with LLM-guided mutation, find prompt sets that dominate a hand-written baseline and single-objective or random search in fidelity, inter-agent disagreement and length?
- **RQ2.** Which component carries the effect: the LLM-guided mutation or the selection scheme?
- **RQ3.** Do the gains hold on unseen events?

1.3 Contributions

1. **Representation and objectives.** An individual is a set of three school genes (diagnosis, concepts, stance, contrast) decoded through the unchanged thesis prompt scaffold. Objectives are school fidelity, inter-school disagreement and prompt length, with Deb's constraint handling on output validity.
2. **Operators.** Structural crossover at school and field level; rule mutation that touches one school per child; and an LLM-guided mutation in which the parent's weakest objective is turned into a rewrite instruction (AI-to-EC).
3. **Protocol.** A written analysis protocol (`code/PROTOCOL.md`, committed before the held-out analysis; deviations in Section 4.8), held-out events from a separate seed, exact sign-flip tests on held-out events with Holm correction, and a fully cached pipeline that can be re-run offline from the committed raw outputs.

2 Related Work

We collected 34 papers from 2024–2026 (`report/references.bib`). Every author list and first-submission date was checked against its arXiv page, and the journal DOIs, volumes and pages against the publisher records in Crossref. The papers are grouped by theme; the classic Promptbreeder and EvoPrompt papers are included because their venue versions are from 2024. The synthesis below is organised by method, not by paper.

2.1 LLMs as evolutionary variation operators

The strongest current trend treats an LLM as the mutation and crossover operator of an evolutionary loop. Promptbreeder evolves task prompts and the mutation prompts themselves (Fernando et al. 2024); EvoPrompt connects LLM-generated crossover and mutation with GA and DE on discrete prompts (Guo et al. 2024); EvoLLM treats an LLM as an in-context recombination step for evolution strategies (Lange, Tian, and Tang 2024). Beyond prompts, LLMs design heuristics and algorithms: EoH co-evolves natural-language thoughts with code (F. Liu et al. 2024), ReEvo adds reflective, verbal feedback to crossover and mutation (Ye et al. 2024), LLaMEA generates metaheuristics that beat CMA-ES on BBOB (Stein and Bäck 2024), and Eureka evolves reward code from simulator feedback (Ma et al. 2024). Automated design of agentic systems applies the same idea to whole agents, with an archive rather than a selection population (S. Hu, Lu, and Clune 2025). Surveys place these methods in a common frame, with the LLM as optimiser, predictor, extractor or designer (X. Wu et al. 2025; F. Liu et al. 2026). Recent work also adds guidance on which parent and which operator to call, to save model calls (Wang et al. 2026).

Gap. Nearly all of this work optimises a single fitness value. Where diversity appears, it is an exploration device inside one search, not an objective.

2.2 Multi-objective and quality-diversity search with LLMs

Pareto-based prompt search is emerging. Pareto Prompt Optimization trains a prompt generator with dominance-based preference losses instead of a fixed scalarisation (Zhao et al. 2025). MOPrompt runs a multi-objective evolutionary search on accuracy versus context length and reports a 31% shorter prompt at the best baseline's accuracy on one sentiment task (Câmara et al. 2025). REMoH wraps NSGA-II around LLM-generated heuristics, with a clustering reflection step that keeps offspring diverse (Forniés-Tabuenca et al. 2025); an LLM-aided MOEA uses an LLM only when a surrogate says it is worth the call (W. Liu, Chen, and Tang 2024). On the quality-diversity side, LLMs can propose archive members in context (Lim, Flageat, and Cully 2024), MAP-Elites explores prompt space by structural descriptors (Santos, Julia, and Nascimento 2025), and CycleQD cycles which objective acts as the quality measure while the others act as behaviour axes (Kuroki et al. 2025).

Gap. Multi-objective LLM search so far targets accuracy versus cost on one task. None uses a *between-agent* distinctness objective, and none evaluates a prompt set on held-out events.

2.3 Diversity and perspective fidelity in multi-agent LLM systems

Multi-agent discussion can increase viewpoint diversity, but conformity and confabulation erode it. Persona-assigned agents drift from their assigned views under debate instructions (Baltaji, Hemmatian, and Varshney 2024). Vanilla debate often underperforms majority vote, and explicitly diversifying the initial viewpoints and calibrating confidence helps (Zhu et al. 2026). In a controlled logic study, group diversity and reasoning strength mattered most, while majority pressure discouraged independent correction (H. Wu, Li, and Li 2025). Diversity collapses across models, not only within one: an open-ended study finds intra-model repetition and inter-model homogeneity (Jiang et al. 2025). For economic reasoning, LLMs lean towards intervention-oriented causal

conclusions, and that bias persists under one-shot prompting (Lee et al. 2026). Macro agents with explicit hawkish and dovish beliefs can improve FOMC classification, but the study does not check whether agents stay faithful to their belief across rounds (Takano, Hirano, and Nakagawa 2025). Survey simulations with LLM agents match macro expectation distributions only when priors are injected (Lin, Sun, and Yan 2025).

Automated multi-agent design is also active. MASS optimises prompts and topology in three stages with a single task score (Zhou et al. 2026). EvoMAS evolves multi-agent configurations by mutation and crossover guided by execution traces (Y. Hu et al. 2026).

Gap. Diversity is studied as an accuracy device or as a property to be measured, but not as an optimisation target next to persona fidelity, and not for competing economic perspectives.

2.4 Evolutionary model merging, fine-tuning and self-evolution

This family is on the course's list of directions and is the closest population-based alternative to our method that changes model weights instead of prompt text. We review it briefly to show where our setting sits.

Evolution has been applied to parameters and recipes rather than text. Evolutionary search over merge recipes, in parameter and data-flow space, yields state-of-the-art Japanese models (Akiba et al. 2025). Multi-objective evolution of merged experts trades cross-task performance against storage (H. Qiu et al. 2026), and a two-dimensional recombination of LLM blocks is searched by bi-level multi-objective evolution (X. Wu, Zhou, and Tan 2026). Layer-wise merging of reasoning-tuned LLMs is another route to combining specialised models (Zhang et al. 2024). Evolution strategies can fine-tune billion-parameter LLMs without gradients (X. Qiu et al. 2026), and black-box evolutionary merging works from API queries alone (Chen et al. 2025). Self-evolving training co-evolves a rubric generator and a policy without human labels (Li et al. 2026).

Gap. These methods optimise weights, which requires training or merging infrastructure. The project's setting, where the only decision variables are the texts of a few prompts, is a cheaper and more interpretable point in the same family.

2.5 Synthesis and positioning

Table 1 summarises the literature along the axes that matter for this project.

Axis	Typical finding	Relevance to our design
LLM as operator	Works for prompts, heuristics and code (Fernando et al. 2024; Guo et al. 2024; F. Liu et al. 2024)	We use an LLM mutation with an objective-directed instruction
Objective	Mostly one scalar; Pareto variants exist for accuracy versus cost (Zhao et al. 2025; Câmara et al. 2025)	Three objectives, one of which is inter-agent distinctness
Diversity	Measured or used as initialisation; homogeneity documented (Jiang et al. 2025; Zhu et al. 2026)	Made an explicit objective

Axis	Typical finding	Relevance to our design
Persona fidelity	Erodes under debate (Baltaji, Hemmatian, and Varshney 2024; H. Wu, Li, and Li 2025)	Measured with a lexicon; held out on separate events
Evaluation	Mostly one task; few held-out checks	Search and held-out seeds are separate; ablations are reported

Our position. To our knowledge no paper combines (i) a *set* of agent prompts as one individual, (ii) a between-agent distinctness objective next to fidelity and length, (iii) an LLM as the mutation operator guided by the weakest objective, and (iv) a held-out check on unseen events. We treat the position as a gap identified from this review rather than as a proof of novelty.

3 Method

3.1 Problem formulation

An individual is a prompt set $P = (g_{\text{kw}}, g_{\text{au}}, g_{\text{mo}})$ with one gene per school. For a set of synthetic events E, each school s answers each event with a parseable report or fails. Three objectives are minimised, all in $[0, 1]$:

- $g_1 = 1 - \text{fidelity}(P)$, where fidelity is the mean lexicon consistency of each school's answers (the thesis consistency score, C_s).
- $g_2 = 1 - \text{disagreement}(P)$, where disagreement is the mean pairwise signal disagreement between the three schools (`mosaic.divergence.distance`).
- $g_3 = \min(1, \text{words}(P)/600)$.

Feasibility requires that at least 90% of school outputs are parseable. Infeasible sets are ranked by their violation, following Deb's rule.

3.2 Genome and decoding

Each school gene has four parts: `diagnosis` (sentences), `concepts` (an ordered subset of a concept pool), `stance` (sentences describing typical readings of events) and an optional `contrast` (sentences of the form "unlike the other schools, you …"). Decoding fills the unchanged `mosaic.divergence.messages()` template, so the output format (causal chain, scenarios, signals) and the events are identical to the thesis harness. The initial population is the hand-written set B0 plus perturbed copies of it.

3.3 Variation

- **Structural crossover** (probability 0.7). With probability 0.5 whole school genes are swapped between parents; otherwise each field of each school is picked from one parent.
- **Rule mutation.** One school is chosen; one or two of: drop or swap a diagnosis sentence, drop, add or reorder a concept. It changes one school, so the other two school outputs are reused from the cache. No rule operator adds a diagnosis sentence or touches `stance` and `contrast`, so rule mutation can only keep or reduce the length of the text it edits (adding a concept adds a

few words). This bias towards shorter sets matters when reading the length results (Section 5.2).

- **LLM-guided mutation** (probability 0.5 of the mutations in the LLM methods). The parent's evaluation picks the objective to improve, with probability proportional to its badness: fidelity (rewrite the least faithful school's diagnosis using its lexicon terms), distinctness (rewrite one field of one of the two most similar schools so that it makes different calls from the other), or length (shorten the longest school's diagnosis to 40 words). The model returns one paragraph; it is accepted if it has 5 to 120 words, and otherwise the child falls back to rule mutation (summed over the three seeds, 8 of 30 LLM children in NSGA-II and 10 of 34 in the GA were rejected this way).

3.4 Selection

- **NSGA-II.** Constrained non-dominated sorting, crowding distance, binary tournament on (rank, crowding), and elitist ($\mu + \lambda$) environmental selection (Deb et al. 2002).
- **GA (B2).** The same operators, but selection by feasibility first and then the equal-weight sum of g_1, g_2, g_3. This isolates the effect of the selection scheme.

3.5 Baselines, ablation and indicator

- **B0.** The hand-written profiles, evaluated with three independent sampling seeds.
- **B1 random search.** Independent random rule perturbations of B0 (1 to 3 operators per school), the same number of evaluations, and the same archive.
- **Ablation.** NSGA-II with rule mutation only (no LLM).
- **Indicator.** Exact 3-D hypervolume of the feasible non-dominated archive, with reference point (1, 1, 1). The archive keeps every feasible set evaluated, so all methods are compared on the same footing.
- **Knee.** The archive member closest to the ideal point (0, 0, 0). The rule was fixed in `code/PROTOCOL.md` before the held-out analysis. The objectives are not rescaled. Across the final fronts g_3 spans about 0.40 to 0.79 and g_2 about 0.50 to 0.78, but g_1 only about 0.47 to 0.60, so fidelity carries the least weight in the distance and the knee tends to be one of the shorter feasible sets.

4 Experimental Setup

4.1 Task and data

The task is a set of synthetic macroeconomic events produced by `mosaic.teacher.make_items`. The search set is the first five events of seed 23 and the held-out set is the first ten of seed 29; one held-out event (`syn-k0007`, seed 29) has the same text as one search event (`syn-k0002`, seed 23), although the two have different macro contexts. Section 5.2 reports the held-out tests with that event removed. Seed 11 (the thesis disagreement matrix) and all thesis held-out files are not used. Lexicons come from `configs/lexicons.toml`, which the thesis marks as PROVISIONAL; this affects the fidelity objective (Section 6).

4.2 Model and backend

Each call goes to an OpenAI-compatible endpoint (`qwen3.8-max`, the model used for the thesis disagreement matrix) with temperature 0.7, top-p 0.9, at most 1,024 output tokens, thinking disabled and a per-run sampling seed. Every response is stored in an append-only cache keyed by the model, the messages and the sampling parameters (`results/cache/llm_cache.jsonl`). A ledger records token counts and an estimated spend, with a USD 30 cap. The spend uses assumed list prices of USD 2.4 and 9.6 per million input and output tokens; the measured token counts are stored, so the cost can be recomputed with the provider's sheet.

4.3 Hyperparameters

Parameter	Value
Population N	6
Generations G	4 (so $N(G + 1) = 30$ evaluations per run)
Search events / held-out events	5 / 10
Algorithm seeds (paired across methods)	1, 2, 3
Crossover / mutation / LLM share	0.7 / 0.9 / 0.5
Tournament size	2
Feasibility threshold	0.9
Length normalisation	600 words
Hypervolume reference	(1, 1, 1)
LLM edit length	5 to 120 words
Sampling	temperature 0.7, top-p 0.9, max 1,024 tokens, thinking off

4.4 Procedure

1. For each method and seed, run the search on the five search events and keep the archive (`results/runs/<method>_s<seed>/gen_log.csv`, `evaluations.jsonl`, `front.jsonl`).
2. Select the knee of the final archive and write it to `results/best_prompts/`.
3. Re-measure each knee, and B0, on the ten held-out events. Each knee is also re-scored from the cache per event (`results/heldout_per_event.csv`).
4. Average over seeds per event and test each method against B0 with an exact paired sign-flip test over the ten events. Correct for eight tests with Holm (`results/heldout_tests.csv`).

4.5 Statistics

Search-level results are mean ± sample standard deviation over three seeds. Held-out intervals are 95% percentile bootstrap intervals over the ten events (10,000 resamples, `mosaic.stats.paired`, seed 0). They are not corrected for multiple comparisons; the p-values are exact sign-flip values and are the ones corrected with Holm. With three paired seeds the smallest attainable one-sided sign-flip p-value is 1/8, so search-level comparisons are reported descriptively. The inferential test uses the ten held-out events, where the smallest attainable two-sided p-value is 2/1024.

4.6 Reproduction

`code/README.md` gives the commands. The matrix was run with `python -m evo_prompt.runner --methods b0 nsga2 ga random nsga2_rule --seeds 1 2 3 --gens 4 --pop 6 --heldout --budget 30`, and the tables with `python -m evo_prompt.analyze` and `python -m evo_prompt.figures`. The analysis reads only the committed cache, so it makes no model calls.

4.7 Computing environment and measured performance

Hardware. The search and the analysis ran on a CPU-only Linux machine (x86_64, no GPU). The language model was accessed over HTTPS through an OpenAI-compatible chat endpoint, so the model's hardware is outside our control.

Software. Python 3.13.18; numpy 2.5.3 (analysis), matplotlib 3.11.2 (figures), pytest 9.2.0 (tests). The model client uses only the Python standard library. The versions are those of the environment in which the reported runs were made; no lock file is provided.

Concurrency. Each batch of model calls sends up to 16 requests in parallel. Every response is written to the cache before the next batch starts.

Measured performance. The figures below come from `results/cache/llm_cache.jsonl`, which stores the response time of every call.

- Recorded calls: 2,849, of which 2,743 belong to the matrix and 106 to the API smoke test.
- Per-call latency: median 14.8 s, mean 14.9 s, 90th percentile 17.0 s, maximum 183.9 s.
- Token totals over all recorded calls: 1,928,385 input and 1,700,982 output tokens.
- Summing the per-call latencies gives about 11.8 hours. This is not wall-clock time, because requests ran in parallel. The wall-clock duration of the matrix run was not logged.
- The 20 offline tests run in under one second (0.7 to 0.9 s across repeated runs).

4.8 Deviations from the protocol

`code/PROTOCOL.md` has not been edited since it was committed, but the analysis departs from it, or from its wording, in three places.

1. **Timing.** The protocol states that it was written before any full-matrix result existed. In fact it was committed (2026-11-01, 15:12 UTC) while the matrix was running, together with a snapshot that already held the seed-1 knee prompt sets of NSGA-II, the GA and random search. The runner measures each knee on the held-out events at the end of its run, so we cannot rule out that some seed-1 held-out values existed at that time. The held-out tables, tests and figures were produced afterwards (16:10 to 16:31 UTC). The protocol was therefore fixed before the held-out *analysis*, not before every result; we do not call it a pre-registration.
2. **Test family.** The protocol applies Holm across method-versus-B0 *and* method-versus-method comparisons (8 + 12 = 20 tests). Tables 3 and 4 use the 8 method-versus-B0 tests. The full 20-test family is in `results/heldout_tests_protocol.csv`: the smallest Holm-adjusted p-value is 0.508, so no conclusion changes. The only method-versus-method comparison with a raw p

below 0.05 is the held-out fidelity of NSGA-II + LLM against the GA (−0.025, raw p = 0.031, adjusted p = 0.56).

3. **Post-hoc sensitivity analysis.** Removing the held-out event whose text duplicates a search event (Section 5.2) was not in the protocol. It was added after the overlap was found.

5 Results and Analysis

5.1 Search level

Table 2 reports the archive and the knee on the search events. Values are mean ± sd over three seeds (`results/summary_table.md`).

Method	Hypervolume	Front size	Knee fidelity	Knee disagreement	Knee validity	Knee words
NSGA-II + LLM	0.1211 ± 0.0182	9.3 ± 1.5	0.461 ± 0.026	0.460 ± 0.025	0.978 ± 0.039	330 ± 50
GA (weighted sum)	0.1220 ± 0.0130	10.7 ± 1.2	0.452 ± 0.038	0.459 ± 0.032	1.000 ± 0.000	305 ± 29
Random search	0.0978 ± 0.0119	4.3 ± 0.6	0.474 ± 0.011	0.437 ± 0.017	1.000 ± 0.000	350 ± 38
NSGA-II, rule only	0.0979 ± 0.0135	7.7 ± 4.0	0.460 ± 0.016	0.429 ± 0.061	0.933 ± 0.000	346 ± 37

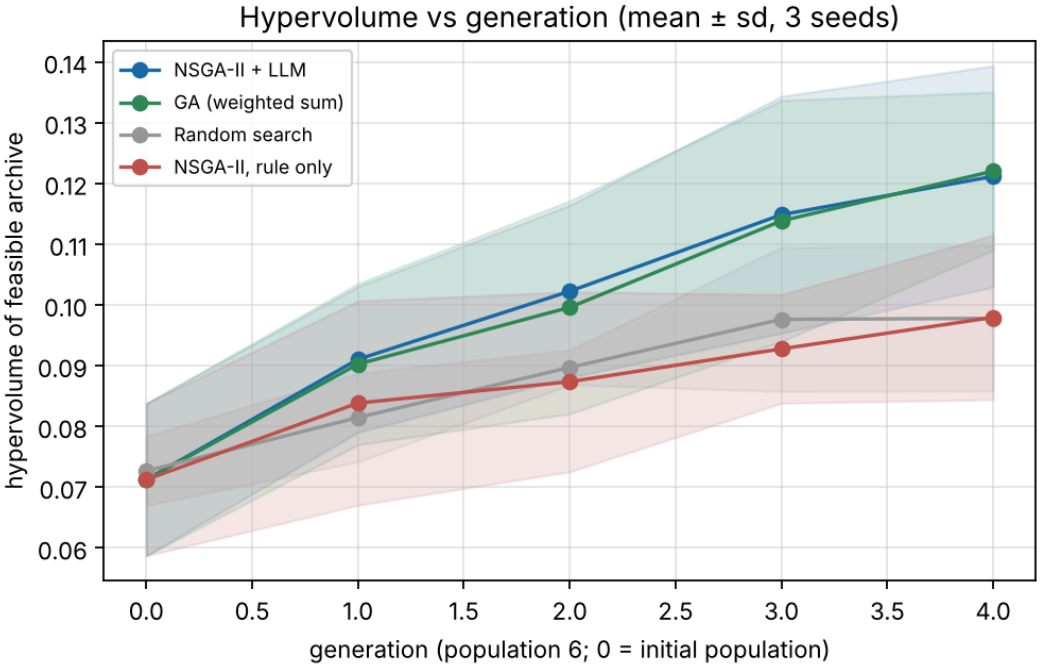


Figure 1. Mean hypervolume of the feasible archive per generation (3 seeds, shaded ±1 sd). Generation 0 is the shared initial population.

Figure 1 shows the mean hypervolume per generation. Generation 0 is the shared initial population (0.071 for NSGA-II, the ablation and the GA alike; 0.073 for random search). From generation 1 on,

NSGA-II and the GA rise above the two methods without LLM mutation (generation 4: 0.121 and 0.122, against 0.098 and 0.098), and the gap widens over the generations.

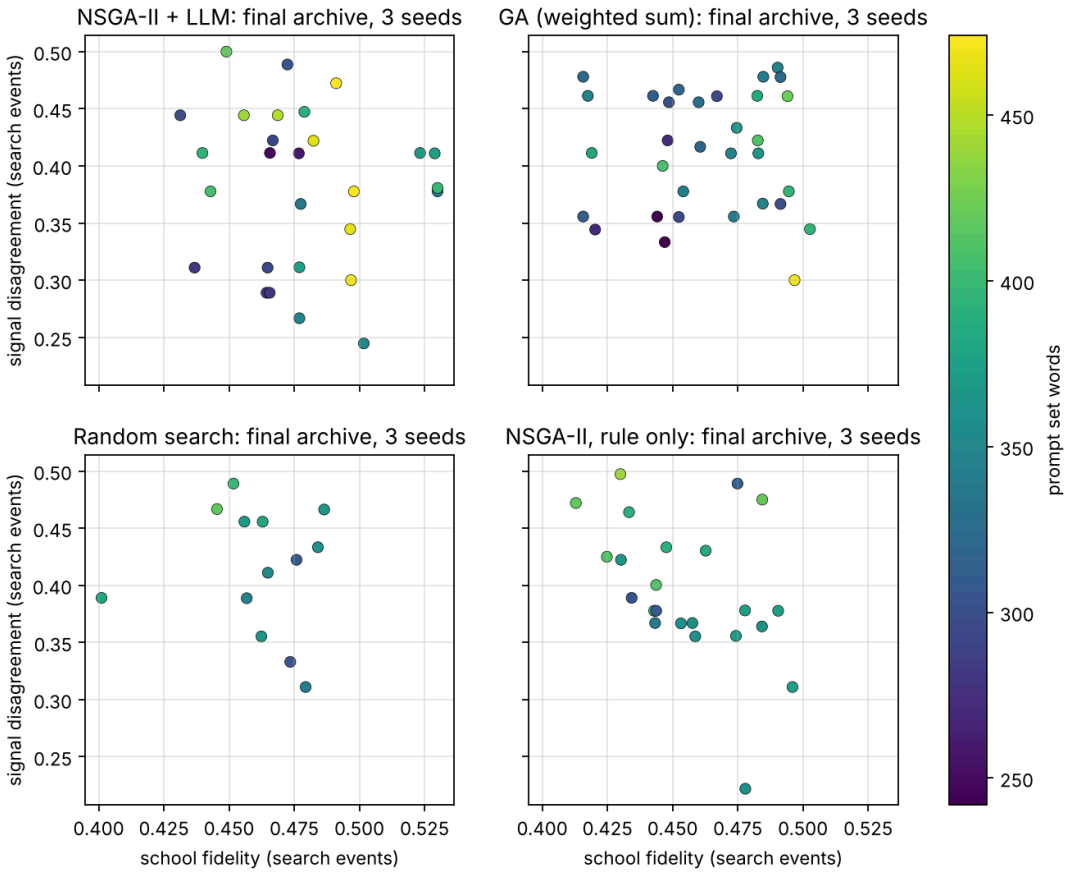


Figure 2. Final feasible archives of the four search methods (3 seeds each, pooled), fidelity against disagreement on the search events, coloured by prompt-set length on a common scale.

Figure 2 shows where the extra hypervolume comes from. Only the two methods with LLM mutation reach sets below about 280 words and fidelity above 0.50 on the search events; the archives of random search and the rule-only ablation stay between about 305 and 440 words, with fidelity at most 0.497.

The pattern suggests an answer to RQ2 at this budget, with three qualifications. NSGA-II without LLM mutation has the same mean hypervolume as random search (0.0979 against 0.0978), so selection alone adds nothing visible over random sampling here. Adding the LLM as mutation operator raises the hypervolume to 0.121 for NSGA-II and 0.122 for the GA. Within each seed, NSGA-II + LLM beats the rule-only ablation in all three pairs (0.114 against 0.087, 0.108 against 0.093, 0.142 against 0.113), the strongest pattern three paired seeds can show (one-sided sign-flip $p = 1/8$). Against random search it is not uniform: with seed 2, random search (0.112) ends above NSGA-II + LLM (0.108).

- **Selection scheme.** The GA's selection is single-objective, yet its archive has the same hypervolume as NSGA-II's. This is weak evidence that selection does not matter. Both methods share the initial population per seed, and after the first generation their hypervolumes are still almost identical (0.0921 for both with seed 2; 0.1025 for both with seed 3); four

generations leave little room for the two selection schemes to diverge. The comparison is also only made with LLM mutation: there is no GA with rule mutation only, so the 2 × 2 design (selection × mutation) is incomplete.

- **Which part of the LLM operator.** The design does not separate the *objective-directed* instruction from LLM rewriting as such. A control in which the LLM rewrites a random field without being told which objective to improve is missing, so the evidence supports "LLM mutation" rather than "objective-guided LLM mutation".
- **Strength.** With three seeds the standard deviations overlap, so all of this is descriptive, not a significance claim.

5.2 Held-out events

Table 3 reports the knee solutions on the ten held-out events. The ± values are not of the same kind: for B0 they are the spread over three sampling seeds of one fixed prompt set, and for the search methods they are the spread over three algorithm seeds, each of which produces a different knee prompt set (and is evaluated with its own sampling seed). The second includes the first.

Method	Fidelity	Disagreement	Validity	Words
B0 hand-written	0.406 ± 0.002	0.407 ± 0.028	0.989 ± 0.019	438
NSGA-II + LLM	0.424 ± 0.017	0.423 ± 0.006	0.989 ± 0.019	330 ± 50
GA (weighted sum)	0.448 ± 0.043	0.419 ± 0.027	0.978 ± 0.019	305 ± 29
Random search	0.432 ± 0.009	0.432 ± 0.024	0.967 ± 0.033	350 ± 38
NSGA-II, rule only	0.433 ± 0.032	0.428 ± 0.048	1.000 ± 0.000	346 ± 37

Table 4 gives the paired tests. Each comparison uses the ten held-out events, and Holm correction covers the eight method-versus-B0 tests.

Comparison	Metric	Mean difference (method − B0)	95% bootstrap CI	p (exact, raw)	p (Holm)
NSGA-II + LLM − B0	fidelity	+0.0179	[-0.0086, +0.0413]	0.2168	1.0000
NSGA-II + LLM − B0	disagreement	+0.0056	[-0.0649, +0.0725]	0.8672	1.0000
GA (weighted sum) − B0	fidelity	+0.0426	[+0.0133, +0.0734]	0.0293	0.2051
GA (weighted sum) − B0	disagreement	-0.0074	[-0.0889, +0.0704]	0.8848	1.0000
Random search − B0	fidelity	+0.0262	[+0.0084, +0.0437]	0.0254	0.2031
Random search − B0	disagreement	+0.0000	[-0.0813, +0.0741]	0.9805	1.0000
NSGA-II, rule only − B0	fidelity	+0.0272	[+0.0041, +0.0503]	0.0566	0.3398

Comparison	Metric	Mean difference (method − B0)	95% bootstrap CI	p (exact, raw)	p (Holm)
NSGA-II, rule only − B0	disagreement	+0.0204	[-0.0518, +0.0852]	0.5664	1.0000

No comparison is significant after Holm correction (smallest adjusted p = 0.203). Before correction, the exact sign-flip p-values for the fidelity gains of the GA and random search are about 0.03. The uncorrected 95% bootstrap intervals exclude zero for the fidelity gains of the GA, random search and the ablation, but the corresponding p-values do not survive the correction across eight tests. We therefore read these fidelity gains as suggestive, not established. Removing the one held-out event whose text duplicates a search event (Section 4.1), the exact sign-flip p-values on the remaining nine events are 0.059 for the GA fidelity gain, 0.051 for random search and 0.109 for the ablation, and no conclusion changes.

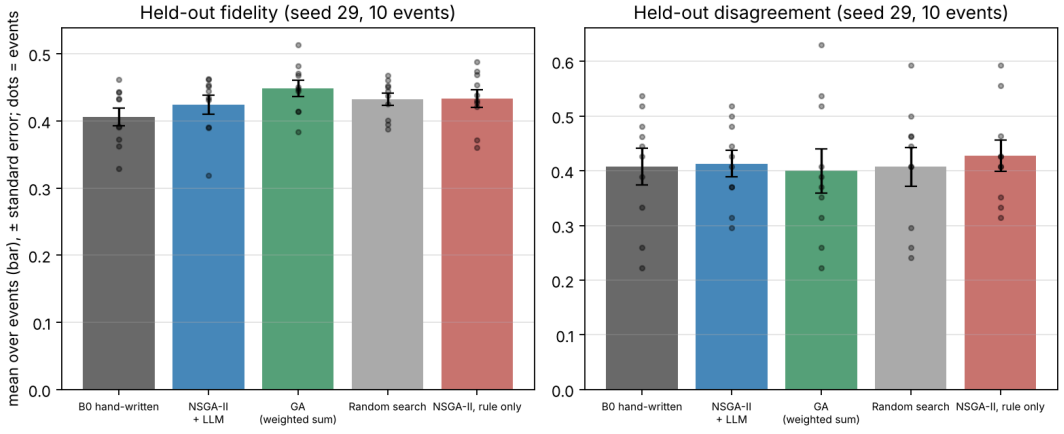


Figure 3. Held-out fidelity and disagreement of B0 and the knee solutions. Bars are the mean over the ten events; error bars are standard errors; dots are individual events.

Three observations follow from the tables.

1. **The prompt sets get shorter, not less faithful.** The searched knees have 305 to 350 words against 438 for B0 (about 20% to 30% shorter), with fidelity and disagreement within a few hundredths of B0. Much of this is built into the design rather than found by the search: rule mutation can only shorten the diagnosis (Section 3.3), and the knee rule weights length more than fidelity (Section 3.5). Random search, which uses only rule mutation and the same knee rule, also ends 20% shorter. The length gain is real for a deployment that pays per token, but it is not evidence that evolution works.
2. **Most of the held-out fidelity gain is not specific to evolution.** Random search gains more fidelity (+0.026) than NSGA-II with LLM mutation (+0.018); the ablation gains +0.027. Random search has no selection *during* the search, but its knee is still the best of 30 sets on the search events by the knee rule, so it is a best-of-30 selection. The gain is therefore better read as an effect of choosing a prompt set on the search events at all than as an effect of evolutionary search. Only the GA's gain (+0.043) is clearly larger, and it is not significant after correction. This is the main caution for the conclusions in Section 6.

3. **Disagreement does not move.** No method changes disagreement by more than 0.021 on held-out events, while B0's own three sampling replicates, with the prompt set held fixed, already range from 0.406 to 0.433 on the same held-out events. The differences in disagreement are inside the sampling noise.
4. **Search-level rankings do not carry over.** On the search events random search has the highest knee fidelity (0.474) and the GA the lowest (0.452); on the held-out events the order of these two is reversed (0.432 against 0.448). With five search events, the knee choice is driven partly by noise, which also limits what the search-level comparison in Section 5.1 can say about generalisation.

5.3 Qualitative observations

Summed over three seeds, 8 of 30 LLM-guided children in NSGA-II (27%) and 10 of 34 in the GA (29%) were rejected by the 5-to-120-word check and replaced by rule mutation (`results/runs/*/evaluations.jsonl`, origin suffix `f`). Reading the accepted edits (the fields of LLM children that do not occur earlier in the same run) shows two kinds of rewrite, which match the length and fidelity instructions.

- **Compression.** Length-directed edits shorten a diagnosis to about 40 words and mostly keep its structure. In the NSGA-II seed-3 knee, the Keynesian diagnosis went from 130 to 40 words: "Assess the output gap via unemployment (natural ~4.5%) and inflation (2% target): negative gaps need support; positive gaps require restraint. Distinguish supply shocks' price versus income effects, noting which dominates. Evaluate policy using the real rate (policy rate minus inflation)." The definitions of the liquidity trap and the paradox of thrift, and the condition on a low policy rate, were dropped. Compression can also change the meaning. In the same run, the Monetarist sentence "a relative-price change becomes inflation only if money accommodates it" was shortened to "Money accommodates supply-shock inflation", which reverses a conditional into a claim. The lexicon-based objectives cannot detect such an error.
- **Lexicon loading.** Fidelity-directed edits rewrite a diagnosis around school vocabulary and grow it to 80 to 110 words. A typical Austrian rewrite (NSGA-II, seed 3) reads "Ask first whether the event alters time preference or initiates credit expansion, for it is artificial credit expansion that distorts interest rates and launches the Austrian business cycle. Such distortion misguides entrepreneurs into lengthening the capital structure … generating widespread malinvestment." These two sentences contain five of the six required Austrian lexicon terms (time preference, credit expansion, Austrian business cycle, capital structure, malinvestment); the B0 diagnosis it replaced contained two (credit expansion, malinvestment). This is the Goodhart risk named in Section 6.2 in its plain form: the edit raises the measured fidelity by naming the school's concepts, and whether it improves the analysis is not measured.

Distinctness-directed edits are harder to recognise in the text. One GA rewrite of the Keynesian diagnosis (seed 2) defines the school against another one ("prioritizing demand management over concerns about capital misallocation"), which is the intended behaviour. The knee prompt sets of every run are in `results/best_prompts/`.

6 Discussion

6.1 What the results say

At this budget, LLM mutation increases the hypervolume of the search archive relative to both random sampling and rule-based NSGA-II in the mean and in every seed against the ablation. This suggests an answer to RQ2 in favour of the LLM as the operator, but it is descriptive (three seeds), it does not isolate the objective-directed instruction, and it does not establish that the selection scheme is irrelevant, since the GA and NSGA-II trajectories are coupled and only four generations long (Section 5.1). On held-out events the searched prompt sets are shorter, with fidelity and disagreement close to the hand-written set, which is a modest answer to RQ1 and RQ3: no dominance over B0 is established, and no method beats random search on the held-out events by a margin that survives correction.

6.2 Limitations

- **Budget.** Population 6 and four generations give 30 evaluations per run. This is a smoke-scale study. It was reduced from the planned population 8 and five generations so that the total stayed under the USD 30 cap (estimated spend USD 20.20).
- **Seeds.** Three seeds per method. The smallest attainable sign-flip p-value on three paired seeds is 0.125, so search-level comparisons are descriptive.
- **Model.** The model is `qwen3.8-max` through an API, not Qwen3-8B on a GPU. Output is cached but not bit-reproducible across providers; the raw responses are committed so that analysis can be repeated without the API.
- **Cost figures.** The USD figures use assumed list prices (USD 2.4 and 9.6 per million tokens), not the provider's invoice. Token counts are measured.
- **Lexicon.** The fidelity objective uses a lexicon the thesis marks as PROVISIONAL. Optimising it invites Goodhart-style gaming: a set can raise its lexicon hits without a better analysis. Expert or model-based judging of faithfulness is the natural fix and is not included.
- **Synthetic events.** The events are generated, and their macro context is synthetic. The results say nothing about real forecasting accuracy.
- **Confounds.** The LLM methods make more model calls per child than rule mutation, and they produce more novel text. The comparison holds the evaluation budget fixed in evaluations, not in calls.
- **Single school family.** The three schools are one economic taxonomy. Generalisation to other perspectives is not tested.
- **Incomplete ablation.** There is no GA with rule mutation only and no LLM mutation without the objective-directed instruction, so the effects of selection and of the guidance itself are not separated (Section 5.1).
- **Knee rule and operator bias.** The knee is chosen by unweighted distance, in which length carries more weight than fidelity, and rule mutation can only shorten the text. Both favour shorter prompt sets independently of search quality (Sections 3.3, 3.5 and 5.2).
- **Protocol timing.** The analysis protocol was committed during the matrix run, not before it (Section 4.8).

6.3 Future work

A full-scale run with population 8 or more, at least five seeds, and a larger search set would give the inferential power the held-out tests lack. The missing arms (a GA with rule mutation only, and an LLM mutation without the objective-directed instruction) would complete the ablation. Replacing the lexicon with a judge model calibrated on human ratings would address the Goodhart risk. The most useful extension is a control that holds the number of model calls fixed across methods, to separate the LLM effect from the extra text it generates.

7 Conclusion

We formulated the design of three competing economic-school agents as a constrained multi-objective search over prompt sets, with fidelity, inter-agent disagreement and length as objectives and an LLM as the guided mutation operator. Under a reduced budget, LLM mutation raised the mean search hypervolume from about 0.098 to about 0.12 and beat the rule-only ablation in every seed, while the two selection schemes, which stayed close for the four generations, reached the same hypervolume. On ten unseen events, the searched prompt sets are 20–30% shorter than the hand-written baseline with similar fidelity and disagreement, but no held-out improvement survives multiple-comparison correction, and random search gains almost as much fidelity as the evolutionary methods. The evidence suggests that LLM mutation is a useful operator for prompt-set search, with the caveats that its fidelity edits partly game the lexicon and its compressions can change meaning. It does not yet support a claim that the evolved sets outperform careful hand-written prompts on unseen events.

Appendix A: Use of AI tools

This appendix follows the course and PolyU guidance on disclosing generative AI; `AI_USE.md` in the submission archive gives the dated log.

- **Literature review.** AI sub-agents, one per theme, searched for 2024–2026 papers and opened the arXiv or venue page of each candidate. Author lists and first-submission dates were checked against arXiv on 2026-11-01, and journal DOIs, volumes and pages against Crossref on 2026-11-02. Venue details come from arXiv comments, venue pages or the course reference list.
- **Code.** The evo_prompt package, tests and runner were written with an AI coding assistant. They pass 20 offline tests (`python -m pytest -q courses/dsai6707/code`) and the repository's own test suite, and an automatic checker (`python -m evo_prompt.check_report`) compares the numbers in this report with the result files.
- **Analysis and figures.** Written with AI assistance; numbers are generated by the scripts in `code/evo_prompt/` and can be regenerated from `results/`.
- **Report prose.** Drafted with AI assistance. The author is responsible for the final wording, the interpretation and the citations.
- **Experiments.** The model calls were made by the code in this repository, through the API, under the budget cap described above.

Appendix B: File map for the numbers

Number	File
One row per run (search and held-out metrics)	`results/summary.csv`
Hypervolume, front size, knee search metrics	`results/summary_search.csv`, `results/summary_table.md`
Held-out means, per-event values	`results/summary_table.md`, `results/heldout_per_event.csv`
Paired tests and Holm	`results/heldout_tests.csv` (8-test family), `results/heldout_tests_protocol.csv` (20-test protocol family)
Hypervolume curves	`results/runs/<method>_s<seed>/gen_log.csv`, `results/figures/hv_vs_generation.png`
Evaluations and origins (LLM children, rejections)	`results/runs/<method>_s<seed>/evaluations.jsonl`
Best prompt sets	`results/best_prompts/<method>_s<seed>_knee.txt`
Raw model outputs	`results/cache/llm_cache.jsonl`
Spend and token totals	`results/logs/matrix.log` (ledger line)
Smoke test	`results/smoke/nsga2_s1/`
Analysis protocol	`code/PROTOCOL.md`

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